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Assignment 2
Due on Feb 25, 2020

One researcher found one organism has a very interesting property. After many steps' purification, he was able to get a small amount of protein. Then he used Mass spectrometry (MS) to find out the sequence and the molecular weight of the protein. The protein sequence is provided in the form of single-letter amino acid code.

1 medaknikkg papfyledg tageqlhkam kryalvpgti aftdahievd ityaeyfems
61 vrlaeamkry glntnhrivv csenslqffm pvlgalfigv avapandiyn erellnsmgi
121 sqptvvfvs kglqkilnvq kklpiiikii imdsktdyqg fqsmytfvts hlppgfneyd
181 fvpesfdrdk tialimnssg stglpkgval phrtacvrfv hardpifgnq iipdtailsv
241 vpfhghgfmf ttglyicgf rvvlmyrfec elfrlsqdy kqsallvpt lfsffakstl
301 idkydlnlh easggapls kevgeavakr fhlpgrqgy gltetsail itpegddkpg
361 avgkvvpffe akvvdldtgk tlgnvqrgel cvrgpmimg yvnnpeatna lidkdgwllhs
421 gdiaywdede hffivdrks likygyqva paelesillq hpnifdagva glpdddagel
481 paavvvlehg ktmtekeivd yvasqvtak klrggvvfvd evpkglgtkl darkireili
541 kakkggkiav

He needs to make more protein to further characterize it. Unfortunately, he does not have enough organisms to provide him enough protein for the characterization.

You need to help him by doing the following things:

- (a) Propose a method to make more protein (as detailed as possible)
- (b) Tell him the molecular weight of this protein
- (c) Convert the protein sequence to corresponding DNA sequence (Please specify which codon you are using)
- (d) What is the putative function of this protein? (using protein blast tools in one of the provided bioinformatics databases)
- (e) Give at least three applications of this protein

Answer:

- (a) & (c): basic strategy: use pET 28(a) platform to express protein. Choose genetically modified *Escherichia coli* (E. Coli) as host cell (with DE3 gene which can express T7 RNA Polymerase). GM E. Coli source is listed at this site: <https://blog.addgene.org/plasmids-101-e-coli-strains-for-protein-expression>. Insert target gene into pET28(a) and transform into E. Coli to get target protein.

Detailed Steps:

1. Convert target protein into DNA sequence (codon usage see below) then purchase DNA sequence with proper cutting site from companies at few hundreds dollars.
2. Purchase genetically modified E. Coli (DE3) as host.
3. Use PCR on DNA to amplify the amount of DNA.
4. Digest enzyme with restriction enzyme to generate proper overhangs. Purify

5. Digest pET28 with enzyme to prepare vectors. Purify
6. DNA ligation with T4 DNA ligase to insert target gene into vectors.
7. Use nuclease to cut vectors and then run gel electrophoresis to make sure DNA is inserted in proper way.
8. When step 7 is confirmed, transform vectors into host cells through electroporation or Heat Shock.
9. Add enough food source and also IPTG to suppress protein expression and let cell concentration to reach maximum value first.
10. When step 9 is done, resuspend in a fresh media to remove IPTG and start protein expression. And also add solution containing Mg ion and ATP to work as cofactor for T7RNAP and protein expression energy source.
11. Add IPTG again after certain amount of time to stop protein expression in case protein amount exceeds cell protein load capacity.
12. Obtain protein through cell lysis. And then collect and isolate target protein from cellular debris.
13. Run protein sequencing to determine whether the protein is of correct sequence or not. Collect protein of correct sequence and abandon rest.

Considering E.Coli codon usage bias, the choice of codon for the target protein is as follows:

atggaagatgcgaaaaacattaaaaaggcccgccgctttatccgctggaagatggc
accgcgggcggaacagctgcataaagcgaatgaaacgctatgcgctggcgccggcaccatt
gcgtttaccgatgcgcataattgaagtggatattacctatgcggaatatttgaaatgagc
gtgcgcctggcggaagcgaatgaaacgctatggcctgaacaccaaccatgcattgtggtg
tgcagcgaaaacagcctgcagttttatgccggtgctggcgcgctgtttattggcgtg
gcggtggcgccggcgaaacgatatttataacgaacgcgaactgctgaacagcatgggcatt
agccagccgaccgtggtgtttgtgagcaaaaaggcctgcagaaaattctgaacgtgcag
aaaaaactgccgattatcagaaaattattattatgtagcaaaaccgattatcagggc
tttcagagcatgtatacctttgtgaccagccatctgcgcgggctttaacgaatatgat
tttgtccggaaaagctttgatcgcgataaaaccattgcgctgattatgaacagcagcggc
agcaccggcctgccgaaaggcgtggcgctgccgcacgcaccgcgtgcgtgcgcttagc
catgcgcgcgacatcgtttttggcaaccagattatccggataccgcgattctgagcgtg
gtgccgtttcatcatggctttggcatgtttaccaccctgggctatctgatttgcggctt
cgcgtggtgctgatgtatcgctttgaagaagaactgtttctgcgcagcctgcaggattat
aaaattcagagcgcgctgctggtgccgaccctgttttagctttttgcgaaaagcaccctg
attgataaatatgatctgagcaacctgcataaattgcgagcggcgccgcccgtgagc
aaagaagtggcggaagcgggtggcgaaacgctttcatctgccgggcattcgccagggtat
ggcctgaccgaaaccaccagcgcgattctgattacccgggaaggcgaatgataaaccgggc
gcggtgggcaaaagtgtgctgtttttgaagcgaagtgtggtgatctggataccggcaaa
accctggcgctgaaccagcgcggcgaaactgtgcgtgcgcggcccgaatgattatgagcggc
tatgtgaacaaccgggaagcgaacacgcgctgattgataaagatggctggctgcatagc
ggcgatattgcgtattgggatgaagatgaacatttttattgtgcatgcctgaaaagc
ctgattaaataaaaggctatcaggtggcgccggcggaactggaagcattctgctgcag
catccgaacatttttgatcgggcggtggcgccctgccggatgatgatgcggcggaactg
ccggcgggcggtggtggtgctggaacatggcaaaaccatgaccgaaaagaaattgtggat

tatgtggcgagccaggtgaccaccgcgaaaaaactgcgcgcgcggtggtgtttgtgat
gaagtgcgaaaggcctgaccggcaaaactggatgcgcgcaaaattcgcgaaattctgatt
aaagcgaaaaaaggcggcaaaattgcggtg

Tools to do this conversion is at: https://www.bioinformatics.org/sms2/rev_trans.html

E.Coli codon usage frequency table is available at:
<https://www.genscript.com/tools/codon-frequency-table>

- (b) : Obtain the data from bioinformatics.org. Molecular weight of this amino acid sequence is 60.65 kilodalton.

https://www.bioinformatics.org/sms/prot_mw.html



(d) & (e): According to the tools available NCBI, there are a few binding sites for this protein.

Database : CDD v3.17-52910 PSSMs

Data in this database indicated a few binding sites of this protein. AMP binding site from 240 to 450 amino acids. From 6 to 440 amino acids, it matches Firefly_Luc and belongs to AFD Class I superfamily.

Firefly_Luc	cd17642	insect luciferase
CaiC	COG3018	Acyl-CoA synthase
PLN02246	PLN02246	CoA Ligase
AMP-Binding	pfam00501	AMP Binding Enzyme
menE	TIGR01923	O-succinylbenzoate-CoA ligase.

From these results, **putative function** of this protein is among first six amino acid residue and last six amino acids.



Three possible usage of this protein:

1. CoA ligase is involved in the oxidation of fatty acid. This protein could be used to treat Hyperlipidemia. (https://en.wikipedia.org/wiki/Long-chain-fatty-acid%E2%80%94CoA_ligase#:~:text=The%20long%20chain%20fatty%20acyl,proceeding%20through%20an%20adenylated%20intermediate.)
2. Acyl-CoA synthase is involved in biosynthesis of fatty acid. The deregulation of cancer cell metabolic network is considered as a promising way to treat cancer. Acyl-CoA synthase could be studied as a starting point to develop cancer therapy. (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6696099/>)
3. AMP is involved in cellular energy homeostasis, largely to activate glucose protein. So it could be used to treat Hyperglycemia. (https://en.wikipedia.org/wiki/AMP-activated_protein_kinase)